

Ronaldo Franco Jaldin

Machine Learning / Research Engineer

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ML research engineer building LLM, retrieval, RL, and computer-vision systems; ICPC World Finalist.

Experience

Freelance Machine Learning Engineer

Bolivia / Remote · Apr 2025–present

- Developing coffee-shop video analytics for product recognition, production counts, customer hand-offs, and time-stamped reports; building the domain dataset with model-assisted annotation.
- Building, since Sep 2026, a WhatsApp interface for Odoo Community Edition that answers natural-language inventory and stock-movement queries; designing tests and evaluating answer quality.
- Used Amazon Bedrock for managed foundation-model inference in freelance AI workflows.
- Delivered a production face-attendance system for a private company using standard cameras, combining face detection with anti-spoofing checks designed to prevent fraudulent check-ins.

Research Assistant — Huawei Research Institute

Moscow, Russia · Mar 2024–Mar 2025

- Built a CodeXGLUE-based embedding pipeline for patch and file retrieval; fine-tuned Jina embeddings with QLoRA and outperformed the off-the-shelf baseline, reaching 0.62 MRR on SWE-bench Verified.
- Optimized prompts and multi-trial sampling for Llama 3.1 issue resolution, obtaining 77% resolution when supplied with oracle-retrieved context.
- Developed agentic and agentless pipelines that create reproducible test environments for arbitrary repositories, and contributed Docker environments to the SWE-bench harness.
- Mined and cleaned issue–patch–file pairs from open-source repositories to supervise retrieval models.

Student Researcher — HSE Laboratory on AI for Computational Biology

Moscow · Mar–Sep 2023

Worked in C++ on parallelization and concurrency optimizations for the Crux proteomics toolkit; studied Siamese networks, contrastive learning, and learned mass-spectrum embeddings.

Research Projects

GFlowNet post-training of BioT5+

BSc thesis · 2025–26

Post-trained BioT5+ with LoRA and Detailed Balance for text-conditioned molecule generation. Compared three replay strategies; supervised references mixed with on-policy rollouts trained most stably.

UniXcoder code retrieval

Python · PyTorch · Transformers

Adapted natural-language code search for Java and Python; explored function/window embeddings, similarity aggregation, and length-based ranking analysis.

Publication

Exploring the Potential of Conversational Test Suite Based Program Repair on SWE-bench

arXiv:2410.04485 · 2024 · Co-authored a study on 92 selected SWE-bench Lite problems; with oracle function localization, test-feedback-driven Llama 3.1 repair passed public and hidden tests for 47% of problems versus 34% for repeated generation.

Education

BSc, Applied Mathematics and Informatics — HSE University

Moscow · 2021–2026

Technical Skills

Languages	Python · C++ · SQL · Bash
ML / LLM / RL	PyTorch · Hugging Face Transformers · PEFT · vLLM · Unsloth · LoRA/QLoRA · Gymnasium · Pandas
Computer vision	OpenCV · YOLO · ByteTrack · ONNX Runtime
Cloud / agents	AWS (Amazon Bedrock) · LangGraph · LlamaIndex · sentence-transformers · FAISS
Methods	LLM post-training · policy optimization · GFlowNets & amortized inference · embedding retrieval · SWE agents
Evaluation	scikit-learn · Weights & Biases
Backend / DevOps	Docker · FastAPI · Pydantic · PostgreSQL · Git/GitHub · GitHub Actions
Integrations	Odoo Community Edition · WhatsApp Cloud API

Service

Scientific Committee — International Olympiad in Informatics (IOI) 2025

Bolivia · 2025

Problem setting, testing and contest design for the international competition.

Honors

- ICPC World Finalist (2022); Topcoder Open South America — 2nd (2019); ICPC South America — 3rd (2020), 10th (2019), 15th (2018); Bolivia Preliminary Contest — 1st (2018–20, 2025).
- IOI participant (2014, 2015, 2017); Ibero-American Olympiad — Bronze (2014, 2015); Bolivian Olympiad in Informatics — Gold (2016), Silver (2013–15).